Discrete Mathematics and its Applications

Exercise 1.8

Prob 48 Continued: . A dominoe covers (i_1+1, j_1) and (i_1+2, j_1). Another dominoe covers (i_1+3, j_1) and (i_1+4, j_1). This process is continued until a dominoe covers $(7, j_1)$ and $(8, j_1)$.

. A dominoe covers (i_2+1, j_2) and (i_2+2, j_2). Another dominoe covers (i_2+3, j_2) and (i_2+4, j_2). This process is continued until a dominoe covers $(7, j_2)$ and $(8, j_2)$.

. Let j_1 be even and j_2 be odd.

$\therefore$ By Defn. 1, $\exists$ integers K_1 and K_2 s.t. $j_1 = 2K_1$ and $j_2 = 2K_2+1$

$(j_2 - j_1) - 1 = (2K_2+1-2K_1) - 1 = 2(K_2 - K_1)$

By defn. 1, $(j_2-j_1)-1$ is even. It's easy to prove that it's even when j_1 is odd and j_2 is even.

. A dominoe covers Row 8, Column j_1+1 and Row 8, Column j_1+2.

. Another dominoe covers Row 8, Column j_1+3 and Row 8, Column j_1+4. This process is continued until a Dominoe covers Row 8, Column j_2-1 and Row 8, Column j_2.

. $i_1 - 1 = 2y - 1 = 2(y-1)+1$. By Defn. 1, (i_1-1) is odd.

. $i_2 - 1 = 2(x+y) - 1 = 2(x+y-1)+1$. By Defn. 1, (i_2-1) is odd.

. A dominoe covers $(2, j_1)$ and $(3, j_1)$. Another Dominoe covers $(4, j_1)$ and $(5, j_1)$. This process is continued until a Dominoe covers (i_1-2, j_1) and (i_1-1, j_1).

. ~~Anot~~ dominoe covers $(2, j_2)$ and $(3, j_2)$. Another Dominoe covers $(4, j_2)$ and $(5, j_2)$. This process is continued until a Dominoe covers (i_2-2, j_2) and (i_2-1, j_2).

. $(j_2-j_1)+1 = (2K_2+1-2K_1)+1 = 2(K_2-K_1+1)$.

If j_1 is odd and j_2 is even, then from Defn. 1, $\exists$ integers K_1 and K_2 s.t. $j_1 = 2K_1+1$ and $j_2 = 2K_2$. $(j_2-j_1)+1 = (2K_2-2K_1-1)+1 = 2(K_2-K_1)$.

. In Both cases, from Defn. 1, $(j_2-j_1)+1$ is even.

. A dominoe covers Row 1, Column j_1 and Row 1, Column j_1+1. Another Dominoe covers Row 1, Column j_1+2 and Row 1, Column j_1+3. This process is continued until a Dominoe covers Row 1, Column j_2-1 and Row 1, Column j_2.